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Dataset Property	What to Measure	Why it Matters	Common Fix
Image Resolution	Width × Height	Controls detail available	Resize, padding
Aspect Ratio	Width/Height	Prevents distortion	Letterboxing, padding
Effective Downsampling	Total stride	Small objects disappear	Reduce pooling, FPN
Color Channels	RGB, Grayscale, RGBA	Wrong input shape	Modify first layer
Intensity Distribution	Histogram, percentiles	Poor gradients	Normalization
Contrast	Bright vs dark regions	Invisible structures	CLAHE, histogram equalization
Noise Level	Gaussian, Salt-Pepper, Sensor noise	Harder learning	Denoising
Blur	Motion or out-of-focus	Removes edges	Deblurring, augmentation
Compression Artifacts	JPEG quality	Fake textures	Augmentation
Class Imbalance	Samples per class	Majority dominates	Focal Loss, weighted sampling

Dataset Property	What to Measure	Why it Matters	Common Fix
Dataset Size	Number of images	Underfitting/overfitting	Transfer learning
Multi-label Statistics	Labels per image	More complex learning	BCE Loss
Label Noise	Wrong annotations	Poor accuracy	Label cleaning
Duplicate Images	Exact/near duplicates	Data leakage	Remove duplicates
Train/Test Leakage	Same subject in train & test	Unrealistic accuracy	Proper split
Subject Distribution	Patients, cameras, locations	Poor generalization	Stratified split
Domain Shift	Different hospitals/cameras	Performance drops	Domain adaptation
Occlusion	Hidden objects	Missed detections	Occlusion augmentation
Object Scale	Small/medium/large	Detection difficulty	Multi-scale training
Bounding Box Statistics	Area, aspect ratio	Anchor design	Custom anchors
Segmentation Mask Quality	Boundary accuracy	Poor segmentation	Mask refinement

Things Missing From Your List

1. Dataset Size ★★★★★

This is probably the most important thing.

Measure:

- Number of images
- Number per class

Risk:

- Small datasets overfit quickly.

Fix:

- Transfer learning
 - Augmentation
 - Cross-validation
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2. Label Quality ★★★★★

Many datasets contain incorrect labels.

Example

Image of a dog labeled as cat.

The model learns wrong information.

Measure

- Random manual inspection
- Label consistency

Fix

- Clean labels
 - Active learning
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3. Data Leakage ★★★★★

Very important.

Example

The same patient appears in both training and testing.

The model memorizes instead of generalizing.

Always check

- Duplicate patients

- Duplicate images
 - Same video frames
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4. Dataset Bias ★★★★★

Example

All healthy scans from Hospital A

All disease scans from Hospital B

The model learns the hospital instead of the disease.

5. Noise ★★★★★

Examples

- MRI noise
- Camera sensor noise
- Rain
- Dust

Need

- Noise analysis
 - Denoising
-

6. Blur ★★★★★

Especially important for

- Medical imaging
 - Face recognition
 - OCR
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7. Duplicate Images ★★★★★

Duplicates inflate accuracy.

Always remove them.

8. Domain Shift ★★★★★

Training

Hospital A

Testing

Hospital B

Performance often drops significantly.

9. Augmentation Suitability ★★★★★

Not every augmentation is appropriate.

For example:

Medical MRI

✗ Random color jitter

Natural images

✓ Color jitter

Satellite images

✗ Vertical flips may be inappropriate depending on the task

10. Annotation Quality ★★★★★

Detection

Bounding boxes

Segmentation

Masks

Classification

Class labels

Poor annotations often limit model performance more than architecture choice.

Missing Evaluation Metrics

Your table focuses on Accuracy and F1.

Different tasks require different metrics.

Classification

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- PR-AUC
- Confusion Matrix

Detection

- mAP
- IoU
- AP50
- AP75
- Recall

Segmentation

- Dice Score
- mIoU
- Pixel Accuracy
- Hausdorff Distance (medical)

Medical Imaging

- Sensitivity
- Specificity
- ROC-AUC
- Cohen's Kappa

Missing Model Design Decisions

The dataset should also influence these choices:

Dataset Characteristic	Model Decision
Small dataset	Transfer learning

Dataset Characteristic	Model Decision
Large dataset	Train from scratch
Tiny objects	Feature Pyramid Network (FPN), higher resolution
Class imbalance	Focal Loss, weighted loss
Noisy labels	Label smoothing, robust loss
Multi-label	Sigmoid + Binary Cross-Entropy
Multi-class	Softmax + Cross-Entropy
Grayscale	1-channel input or replicated channels
Limited GPU memory	Smaller backbone, mixed precision
Large images	Patch-based training or tiling

If this is for research

A stronger diagnostic pipeline is:

1. Dataset Audit

- Size
- Resolution
- Channels
- Labels
- Class balance
- Duplicates
- Leakage

2. Image Statistics

- Mean
- Standard deviation
- Histograms
- Contrast
- Noise
- Blur

3. Label Statistics

- Class counts
- Multi-label frequency

- Co-occurrence matrix

4. Geometry Analysis

- Resolution
- Aspect ratio
- Object size distribution
- Downsampling effects

5. Domain Analysis

- Camera/device
- Hospital/source
- Subject demographics (when relevant and appropriate)
- Domain shift

6. Model Decisions

- Architecture
- Loss function
- Input size
- Augmentations
- Optimizer
- Learning rate schedule

7. Validation

- Task-specific metrics
- Error analysis
- Per-class performance
- Calibration (for probabilistic models)

This expanded framework is closer to what experienced ML engineers and researchers use when assessing a new dataset before selecting or designing a model.